A.6.1 Incorporating prior knowledge into entropy search

This exercise, implemented in `CH06/02 - Exercise 1.ipynb`, shows us an instance of using different priors when finding the information-theoretically optimal decision, and ultimately will help us further appreciate the elegance and flexibility of entropy search as a generic decision-making-under-uncertainty procedure.

1. Prove that $Pr(X = 1) + Pr(X = 2) + \ldots + Pr(X = 10) = 1$.
   We can do this by simply adding the probabilities together:
   $1 / 2 + 1 / 4 + \ldots + 1 / 2^9 + 1 / 2^9 = 1 / 2 + 1 / 4 + \ldots + 1 / 2^8 + 1 / 2^8 = \ldots = 1 / 2 + 1 / 2 = 1$.
2. Calculate the entropy of this prior distribution.
   Remember the formula for the entropy is $p \log p$. We can write a Python function that computes this sum:

```
def compute_entropy(first, last):
    entropy = 0
    for i in range(first, last + 1):
        p = marginal_probability(i, first, last) ❶
        entropy += -p * np.log2(p) ❷

    return entropy
```

   ❶ Get the current probability.

   ❷ Sum over the terms.

   This function takes `first` and `last` as parameters, which correspond to the smallest and biggest numbers X could be (which start out as 1 and 10), respectively. We then iterate through the numbers between `first` and `last` and add up the $-p \log p$ terms. Here, `marginal_probability()` is a helper function that computes $Pr(X = n)$, which we implement as:

```
def marginal_probability(floor, first, last):
    if floor == last: ❶
        return 2 ** -(last - first) ❶

    return 2 ** -(floor - first + 1)
```

   ❶ Edge case when the floor is the highest possible floor.

   Run `compute_entropy(1, 10)` gives us 1.99609375. This is the entropy of the prior distribution for X.
3. Given the prior distribution defined between 1 and 10, what is the probability that the phone will break when dropped from the 2nd floor? What is this probability for the 5th floor? The 1st floor?
   The probability that the phone will break when dropped from the 2nd floor is exactly $Pr(X = 1)$, which is 0.5.
   The probability that the phone will break from the 5th floor is the probability that $X \geq 4$, which is $Pr(X = 1) + Pr(X = 2) + Pr(X = 3) + Pr(X = 4) = 15 / 16 = 0.9375$.
   The two calculations above could be implemented as a function:

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